

DSA06 – DATA HANDLING AND VISUALIZATION

Activity: Identify Skewness using Violin Plots | Slot C

R.No: 24	Reg. No: 192325078	Name: N. HARSHITH
Department: AIML	Course: Data Handling and Visualization	

Part A – Read the four violin plots

A longer tail toward higher values indicates right (positive) skew, so mean is usually greater than median. A longer tail toward lower values indicates left (negative) skew, so mean is usually less than median. A roughly balanced shape suggests mean $\approx$ median.

Chart 1: Rainfall (mm)

Violin	Skewness	Mean vs. Median
Station 3	Right-skewed	Mean > Median
Station 1	Left-skewed	Mean < Median
Station 5	Left-skewed	Mean < Median

Chart 2: Resting heart rate (beats per min)

Violin	Skewness	Mean vs. Median
Club D	Left-skewed	Mean < Median
Club B	Right-skewed	Mean > Median
Club A	Approximately symmetric	Mean $\approx$ Median

Chart 3: Order value (Rs hundred)

Violin	Skewness	Mean vs. Median
Store 5	Right-skewed	Mean > Median
Store 4	Left-skewed	Mean < Median
Store 3	Approximately symmetric	Mean $\approx$ Median

Chart 4: Battery life (hours)

Violin	Skewness	Mean vs. Median
Model T	Left-skewed	Mean < Median
Model R	Approximately symmetric	Mean $\approx$ Median
Model S	Right-skewed	Mean > Median

Part B – Short answers

B1. Club D has a tail extending toward lower heart-rate values, so it is left-skewed (negative skew). The **median** is the better measure of a typical resting heart rate because it is less affected by the long tail and unusually low values than the mean.

B2. ggplot2 code to draw violins with the median shown

```
library(ggplot2)

ggplot(df, aes(x = group, y = value)) +
  geom_violin(trim = FALSE) +
  stat_summary(fun = median, geom = "point",
    shape = 21, size = 2.5,
    fill = "white", colour = "black") +
  labs(x = "Group", y = "Value") +
  theme_minimal()
```

This assumes df already contains columns named group and value, as stated in the question.

Result: Skewness and the expected mean–median relationship were identified for each violin. The ggplot2 code shows the median as a white point.